Communication Systems Lecture-15: Detector Circuits

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Envelope Detector (1)

- An ideal Envelope Detector is a circuit that produces at its output a waveform that is proportional to the real envelope $R(t)$ of its input passband signal.
- The bandpass input can be written as:

$$v_{in}(t) = R(t)\cos\left[\omega_c t + \theta(t)\right]$$

where $R(t) \geq 0$

- In this case, the output of the ideal envelope detector must be:

$$v_{out}(t) = KR(t)$$

where K is a proportionality constant.

Envelope Detector (2)

- The envelope detector is typically used to detect AM signals.
- In this case:

$$v_{in}(t) = \mathrm{Re}\left\{g(t)e^{j\omega_c t}\right\}$$

where the complex envelope of $v_{in}(t)$ is given by:

$$g(t) = A_c\left[1 + m(t)\right]$$

where $A_c > 0$ and $m(t)$ is the modulating signal.

- The output of the envelope detector is:

$$v_{out}(t) = KR(t) = K|g(t)| = KA_c|1 + m(t)|$$

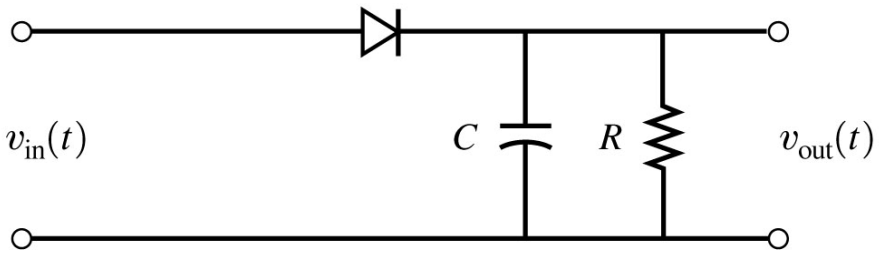
- If $|m(t)| < 1$, then:

$$v_{out}(t) = KA_c\left[1 + m(t)\right] = KA_c + KA_c m(t)$$

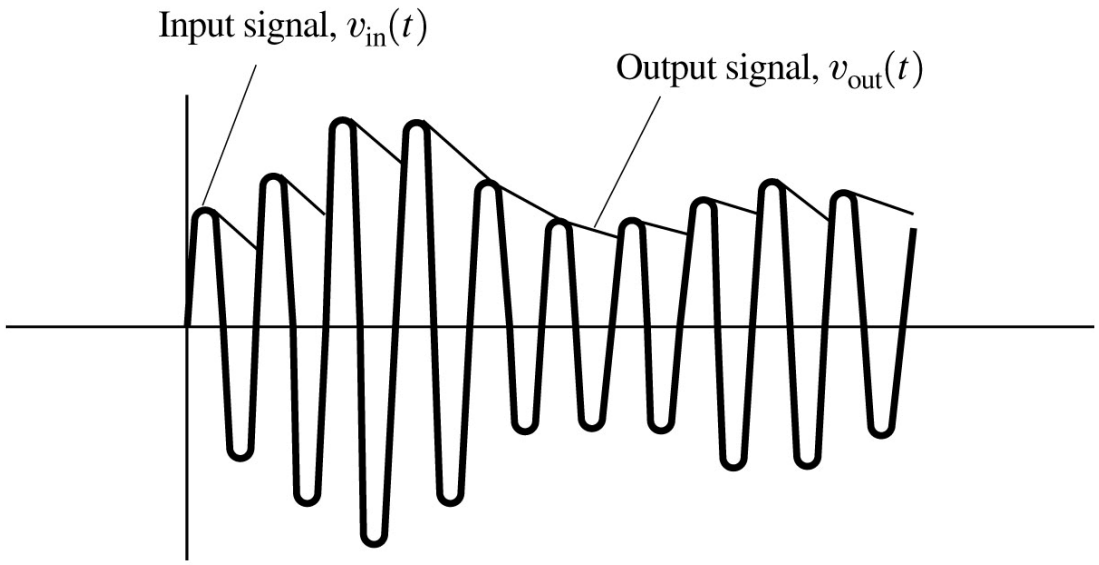
  - KA_c is the DC voltage. It is used to provide automatic gain control (AGC) for the AM receiver. In other words, if KA_c is relatively small (a weak AM signal is received), the receiver gain is increased and vice versa.
  - $KA_c m(t)$ is the detected modulation.

Envelope Detector (3)

- The next figure shows a simple diode detector circuit that approximates an ideal envelope detector.



(a) A Diode Envelope Detector



(b) Waveforms Associated with the Diode Envelope Detector

Envelope Detector (4)

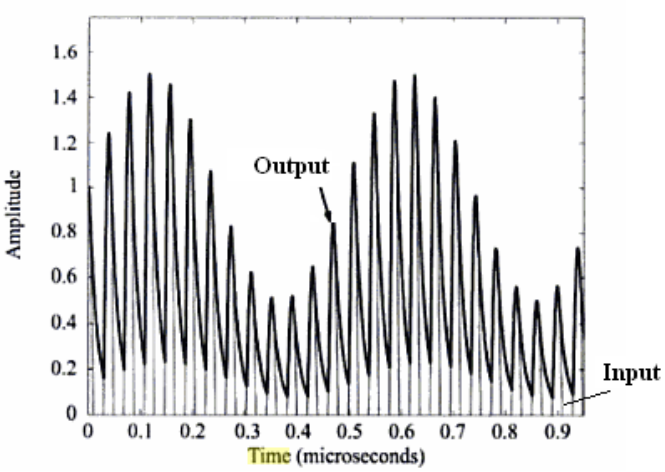
- The diode current occurs in pulses that are proportional to the positive part of the input signal. The current pulses charge the capacitor that will later discharge during the negative part of the input signal to produce the output signal illustrated in the previous figure.
- The RC time constant must be chosen to verify the relation:

$$B \ll \frac{1}{2\pi RC} \ll f_c$$

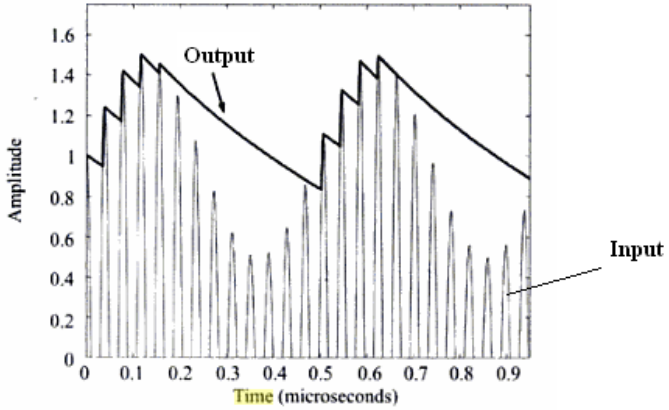
  where RC is the time constant of the filter.
- The last relation assures that the output signal follows the real envelope $R(t)$ and not the rapid variations in $v_{in}(t)$.

Envelope Detector (5)

- If $\frac{1}{2\pi RC} > f_c$ (RC is very small), the detector will follow the rapid variations in $v_{in}(t)$ (not the real envelope):

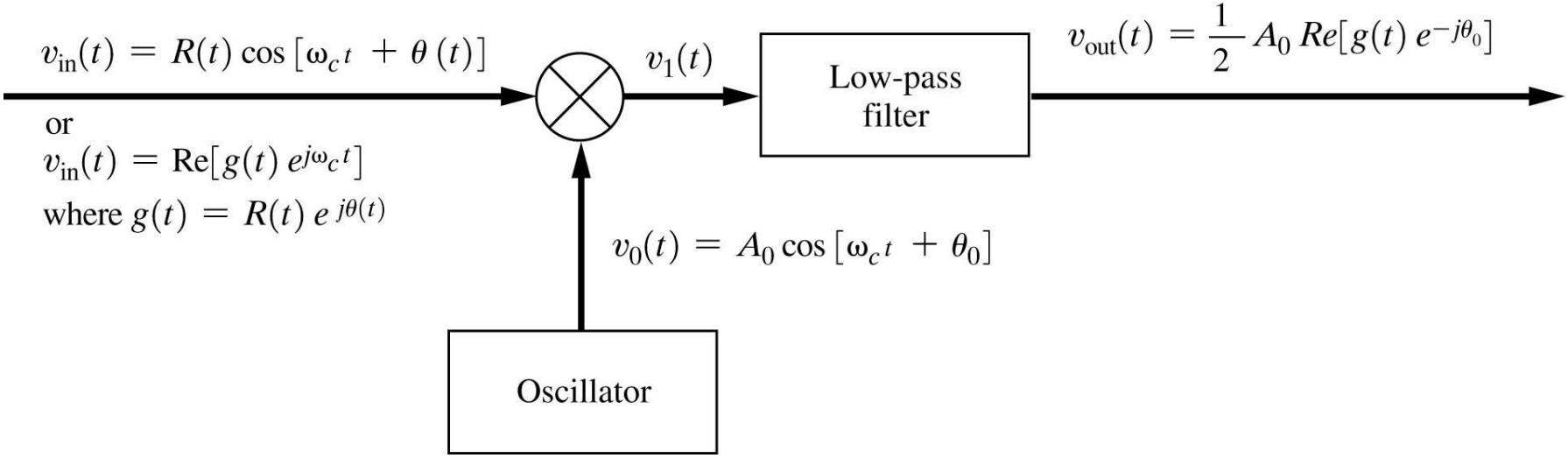


- If $\frac{1}{2\pi RC} < B$ (RC is very large), the detector is slower than the real envelope:



Product Detector (1)

- The Product Detector is a mixer circuit that down-converts the input passband signal to baseband.



Product Detector (2)

- The output of the multiplier is:

$$
\begin{aligned}
v_1(t) &= R(t)\cos\left[\omega_c t + \theta(t)\right] A_0 \cos\left(\omega_c t + \theta_0\right) \\
&= \frac{1}{2} A_0 R(t) \cos\left[\theta(t) - \theta_0\right] + \frac{1}{2} A_0 R(t) \cos\left[2\omega_c t + \theta(t) + \theta_0\right]
\end{aligned}
$$

where the frequency of the oscillator is f_c and its phase is θ_0.

- The signal at the output of the lowpass filter is:

$$
\begin{aligned}
v_{out}(t) &= \frac{1}{2} A_0 R(t) \cos\left[\theta(t) - \theta_0\right] \\
&= \frac{1}{2} A_0 \mathrm{Re}\left\{ g(t) e^{-j\theta_0} \right\}
\end{aligned}
$$

where $g(t)$ is the complex envelope of the input:

$$
g(t) = R(t) e^{j\theta(t)} = x(t) + jy(t)
$$

where $x(t)$ and $y(t)$ are the quadrature components.

Product Detector (3)

- Because the frequency of the oscillator is the same as the carrier frequency of the incoming signal, the oscillator has been frequency synchronized with the input signal.
- Furthermore, if, in addition, $\theta_0 = 0$, the oscillator is said to be phase synchronized with the in-phase component ($x(t)$).
- If $\theta_0 = 0$:

$$v_{out}(t) = \frac{1}{2}A_0 \mathrm{Re}\left\{g(t)\right\} = \frac{1}{2}A_0 x(t)$$

  the output is proportional to the in-phase component $x(t)$.
- If $\theta_0 = \pi/2$:

$$v_{out}(t) = \frac{1}{2}A_0 R(t)\cos\left[\theta(t) - \pi/2\right] = \frac{1}{2}A_0 R(t)\sin\left[\theta(t)\right] = \frac{1}{2}A_0 y(t)$$

  the output is proportional to the quadrature component $y(t)$.

Product Detector (4)

- The product detector is sensitive to both AM and angle modulation.
- Suppose that the input contains no angle modulation: $\theta(t) = 0$.
- In this case, if the reference phase is set to zero ($\theta_0 = 0$), then:

$$v_{out}(t) = \frac{1}{2} A_0 R(t)$$

- In this case, the real envelope is obtained at the output of the product detector.
- Note that while the envelope detector is nonlinear, the product detector is linear.

Product Detector (5)

- Suppose that the input contains no amplitude modulation: $R(t) = A_c$.
- If the reference phase is $\theta_0 = \pi/2$, then:

$$v_{out}(t) = \frac{1}{2}A_0 A_c \cos[\theta(t) - \pi/2] = \frac{1}{2}A_0 A_c \sin[\theta(t)]$$

- Assuming that the phase difference is small, $|\theta(t)| \ll \pi/2$, then $\sin\theta(t) \approx \theta(t)$ and consequently:

$$v_{out}(t) = \frac{1}{2}A_0 A_c \theta(t)$$

- In this case, the product detector corresponds to a phase detector. The output of the product detector is proportional to the phase difference (when the phase difference is small).

Frequency Modulation Detector (1)

- An ideal Frequency Modulation (FM) detector is a device that produces an output that is proportional to the instantaneous frequency of the input.
- If the bandpass signal at the input of the detector is given by $v_{in}(t) = R(t)\cos[\omega_c t + \theta(t)]$, then the output of the ideal FM detector is:

$$v_{out}(t) = K\frac{\mathrm{d}\left[\omega_c t + \theta(t)\right]}{\mathrm{d}t} = K\left[\omega_c + \frac{\mathrm{d}\theta(t)}{\mathrm{d}t}\right]$$

  where K is a proportionality factor.
- Usually, FM detectors are balanced. This means that the DC voltage $K\omega_c$ does not appear at the output if the detector is tuned to the carrier frequency f_c. In this case, the output is:

$$v_{out}(t) = K\frac{\mathrm{d}\theta(t)}{\mathrm{d}t}$$

Frequency Modulation Detector (2)

- A FM-modulated signal is given by:

$$s(t) = A_c \cos\left[\omega_c t + \theta(t)\right]$$

where:

$$\theta(t) = K_f \int_{-\infty}^{t} m(t_1)\mathrm{d}t_1$$

and K_f is a proportionality factor.

- In other words, the instantaneous frequency of the modulated signal is proportional to the information signal $m(t)$ since:

$$\frac{\mathrm{d}\theta(t)}{\mathrm{d}t} = K_f m(t)$$

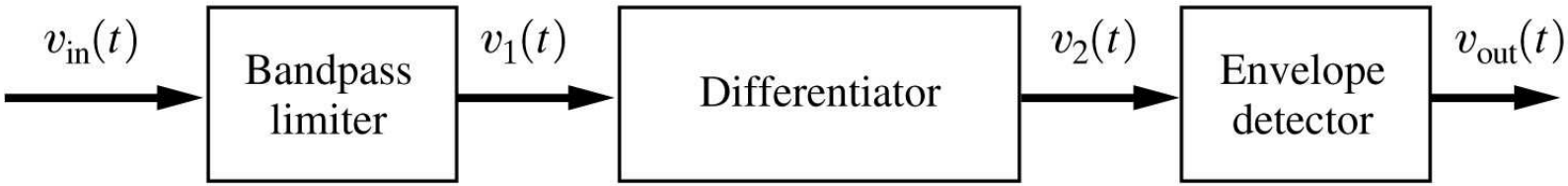
- At the output of the communication channel (and the input of the receiver), the signal can be written as:

$$v_{in}(t) = A(t)s(t) = A_c A(t) \cos\left[\omega_c t + \theta(t)\right]$$

where $A(t)$ stands for the amplitude variations (fading) introduced by the channel.

Frequency Modulation Detector (3)

- A slope detector is one example of a FM detector.



- A bandpass limiter is needed to suppress the undesired amplitude variations ($A(t)$) on the input signal. The output of the limiter is proportional to:

$$v_1(t) = V_L \cos\left[\omega_c t + \theta(t)\right]$$

- The output of the differentiator is given by:

$$v_2(t) = \frac{\mathrm{d}v_1(t)}{\mathrm{d}t} = -V_L\left[\omega_c + \frac{\mathrm{d}\theta(t)}{\mathrm{d}t}\right]\sin\left[\omega_c t + \theta(t)\right]$$

Frequency Modulation Detector (4)

- The output of the envelope detector is the magnitude of the complex envelope of $v_2(t)$:

$$v_{out}(t) = \left| -V_L \left[\omega_c + \frac{\mathrm{d}\theta(t)}{\mathrm{d}t} \right] \right|$$

- In practice $\omega_c \gg \mathrm{d}\theta(t)/\mathrm{d}t$. Consequently:

$$\begin{aligned} v_{out}(t) &= V_L \left[\omega_c + \frac{\mathrm{d}\theta(t)}{\mathrm{d}t} \right] \\ &= V_L \omega_c + V_L K_f m(t) \end{aligned}$$

- The output consists of:
  - A DC voltage $V_L \omega_c$.
  - An AC voltage $V_L K_f m(t)$ that is proportional to the modulation on the FM signal.

Coherent and Noncoherent Detectors

Detectors can be classified as being either coherent or noncoherent.

- Coherent Detectors: they have two inputs:
  - One for a reference signal such as a synchronized oscillator signal.
  - One for the modulated signal that is to be demodulated.
- Noncoherent Detectors: they have only one input that corresponds to the signal to be detected.
- For example, product detectors are coherent detectors while envelope detectors and slope detectors are noncoherent detectors.